GOOGLE CLOUD AI BUILDER CUP 2026 · RETAIL & COMMERCE

Taal

Taal turns a write-off forecast into a human-approved play (a transfer, an offer or both). Its agent is not allowed to act until it has designed a test of itself, and it tells you plainly when the test has not proved anything yet.

₹9,191 transfer plus targeted nudge: projected recovery vs doing nothing on one tea lot PROJECTED

The agent that sells what the forecast says you'll throw away.

TEAM

Nandish · lead

Rinu · co-builder

PROBLEM

1. Forecasts and offers live in different tools: one dark store writes stock off while the next runs out.
2. Customers get blanket markdowns, not offers matched to what they buy.
3. Nothing plans to the online sell-by cut-off: 30% of shelf life or 45 days at delivery.[1]

[1] FSSAI advisory to e-commerce FBOs, 3 Dec 2024, file RCD-13/1/2024-Regulatory-FSSAI(E-13150): minimum 30% shelf life or 45 days at delivery. Either-limit reading is Taal's configurable default.

THE PROBLEM · HOW BIG, WHICH RULE, WHO DECIDES

The forecast sees the write-off coming, but nothing acts on it.

HOW BIG · AS REPORTED

1.8%

of net order value (NOV) is lost to inventory at Blinkit.

As reported by Inc42, 22 Jul 2026. Per-store line below shows its inputs, from the same article.

₹17,132 cr ÷ 2,443 stores × 4 × 1.8% ≈ ₹0.5 cr per store a year

All inventory loss, one quarter annualised. Context for scale, not a Taal result.

THE RULE · AN ADVISORY, NOT LAW

30% or 45 days

of shelf life must remain at delivery of an online order.

FSSAI advisory to e-commerce food sellers, 3 Dec 2024. Stock can be in date on the shelf yet past its online sell-by, so the deadline can come weeks before expiry.

Taal plans to that online sell-by line, not the printed expiry. Either-limit reading is its configurable default.

WHO IT IS FOR

A category or supply-chain head

They own write-off and are the buyer. Day to day, two people use Taal:

A demand planner checks the gap and the guardrails, then approves the play on the Play Desk.

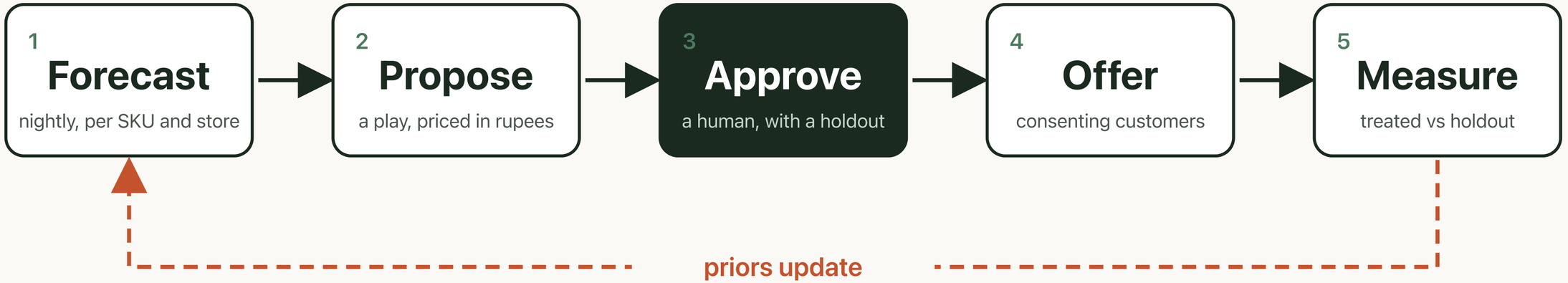
A dark-store manager confirms what is on the pallet on the phone view and receives any transfer.

NEXT The loop and the hero result are on slide 3; projected impact is on slide 15; 3 of 7 practitioners we asked write off weekly or daily (slide 13).

THE LOOP AND THE HERO PLAY

Transfer plus targeted nudge: ₹9,191 vs doing nothing on one tea lot, ₹6,186 ahead of a blanket markdown

PROJECTED · SEEDED TENANT



TEA · GEMINI RUN · PROJECTED

₹9,191

transfer plus targeted nudge (₹9,397 waste avoided – ₹206 transfer); ₹6,186 ahead of a blanket 20% markdown

HOLDOUT · SYNTHETIC ORDERS

+0.3 pp

pp = percentage points. 95% confidence interval −8.7 to +1.6. 1 of 355 treated responded, 0 of 39 held out. SYNTHETIC, inconclusive. A 5 pp lift needs about 1,156 customers (SIMULATION)

WHAT THIS PROVES

The loop closes

Measure runs end to end on synthetic orders; effect size needs a real pilot

THEME Taal forecasts stock per SKU and node nightly, with festival effects, and drafts human-approved offers.

WHAT IS NEW · DIFFERENTIATION

The agent is refused until it designs its own test.

	ONLINE SELL-BY CUT-OFF	HOLDOUT PER PROMO	PROMO → FORECAST WRITE-BACK	INDIA / Q-COMMERCE
Taal	Yes	Yes	Yes	Built for it; seeded tenant
Blue Yonder	Not verified	Not verified	Not verified	Partial
RELEX	Partial	Not verified	Partial	Partial
Wasteless	No	Not verified	Not verified	Not verified
Afresh	No	Not verified	Not verified	No
Flashfood	No	No	No	No

Not verified = no public evidence found in vendor pages read on 2026-10-09; it does not mean the feature is absent. Partial and No are our reading of those pages. Afresh and Flashfood are different product categories.

1 · Gemini proposes

mechanic, audience and rationale for the at-risk lot, in an ADK loop

2 · Code refuses it without a test

The rule (rule_holdout_required): reject if the holdout fraction is outside [0.05, 0.9], min_treated_n is unset, or the seed is missing

3 · The result says what it proved

Tea: +0.3 pp, 95% CI −8.7 to +1.6, INCONCLUSIVE, SYNTHETIC. About 1,156 customers needed for a 5 pp lift (SIMULATION)

Each safety layer measured on its own: the tool refused every cross-customer call (the model tried 3 of 3 times); Model Armor blocked 3 of 8 chat injection cases and fails open; V6 is open. **Method:** protocol committed before any model call, a run is INVALID if a fallback model answers, spend cap enforced.

LIST OF FEATURES

Six capabilities, gated by rules a model never touches.

Vision intake

Photograph a pallet; dates and counts come back with confidence. Low confidence goes to a person.

Nightly Sense

Per-SKU, per-node forecasts. Six gap types, each with rupees and a deadline.

Planner Agent

ADK LoopAgent, max 3 iterations, six deterministic tools; a Cost Governor picks the gaps.

Approve + assignment

A human approves. Holdout by hash; offers go to treated customers only.

Customer Agent

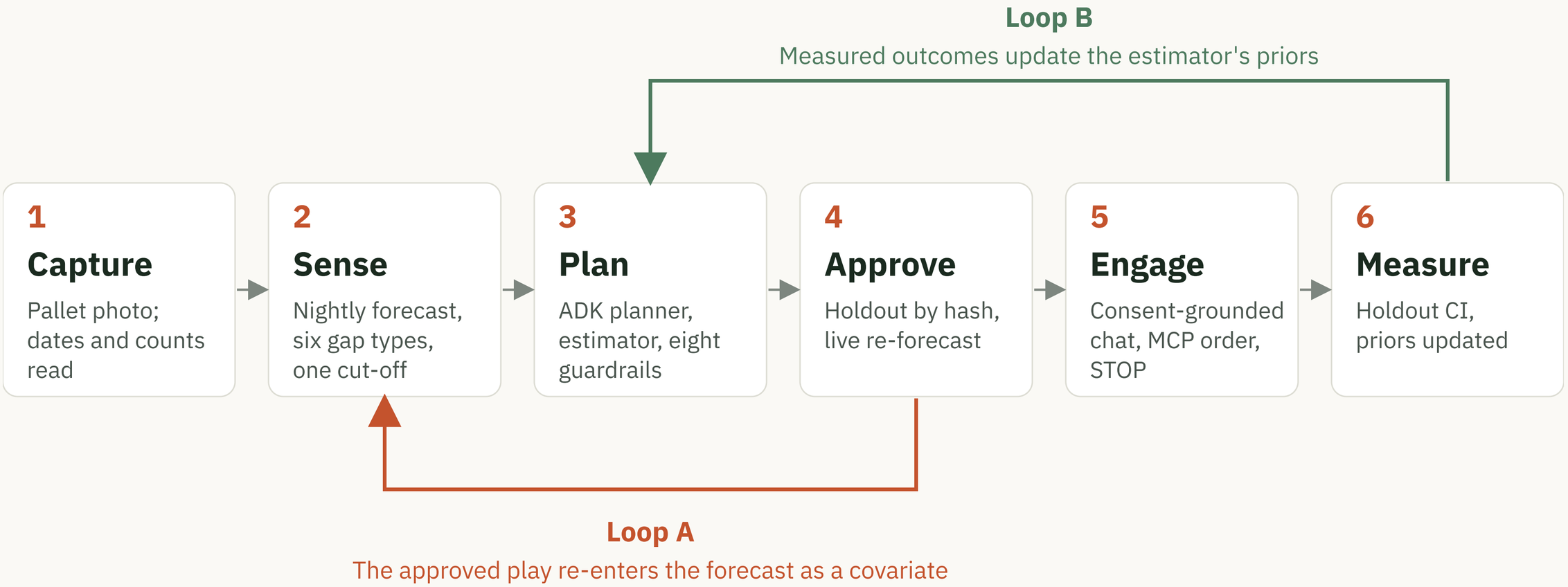
Grounded in stock, offers and consent. Orders through an in-process MCP tool.

Measure

Treated vs holdout, 95% CI. Priors update from the counts.

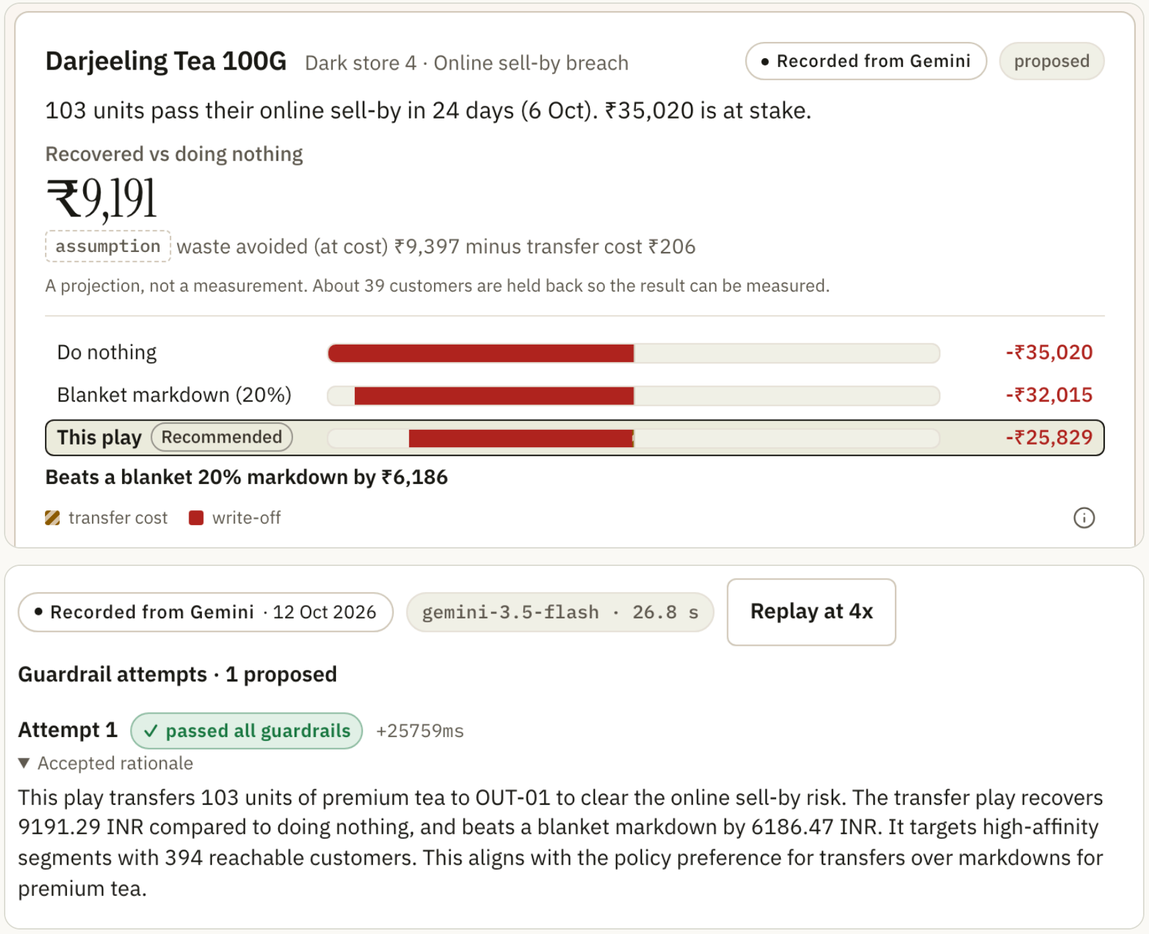
PROCESS FLOW

Two closed loops, not an open-ended pipeline.



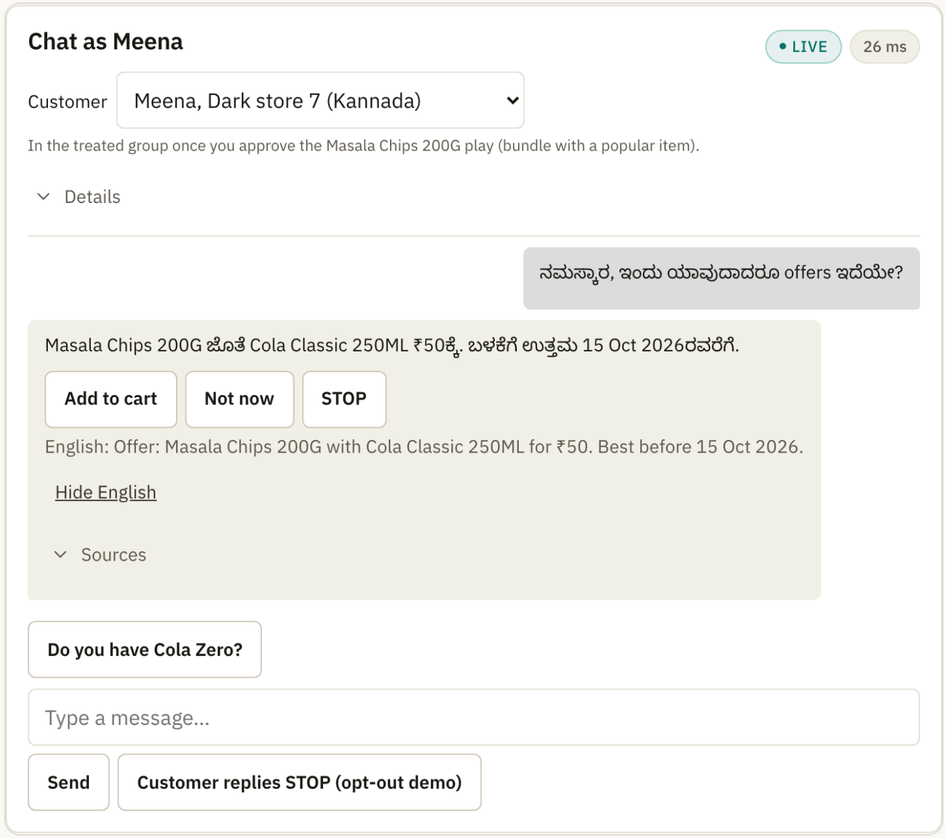
SCREENS

Gemini drafts the play; code prices every rupee; customers get Kannada plus English.



Play Desk: the tea play, recorded from Gemini, 12 Oct

Gemini drafted the play and its rationale; code priced every rupee and the guardrails passed it on the first attempt. Projected, seeded tenant.

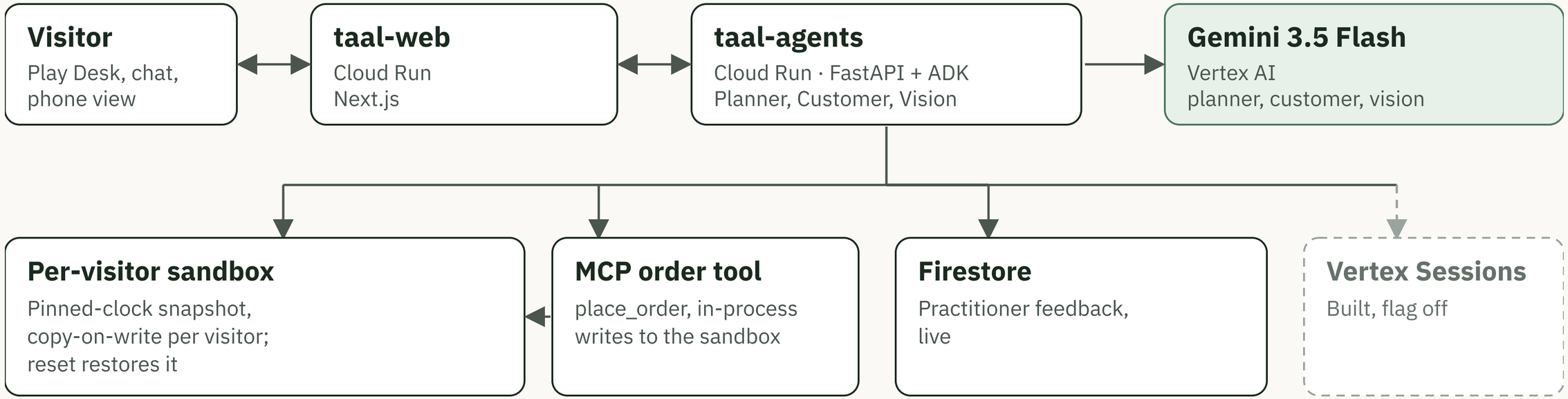


Customer chat, live

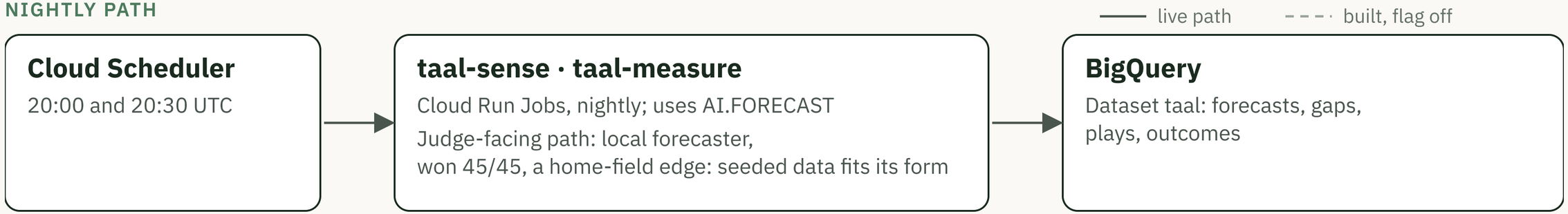
Kannada reply with an English gloss; the order goes through the MCP order tool.

ARCHITECTURE

A sandbox serves judges; nightly jobs run on BigQuery.



NIGHTLY PATH



TECHNOLOGIES USED

Each Google service, and the decision it makes.

SERVICE	THE DECISION IT MAKES
Google ADK planner (Gemini 3.5 Flash)	Proposes the play: mechanic, audience, window, in a LoopAgent (max 3) with 6 tools
Gemini on Vertex AI	Drafts the offer, chats with the customer, reads the pallet photo
Model Armor	Screens every chat message and reply for injection and jailbreak (live 14 Oct); fails open, code guardrails still apply
BigQuery AI.GENERATE_TABLE	Pre-generated the offer copy for the 7 seeded plays; stored and reused at Approve. Other plays try one call within 3 s, then templates.
Local forecaster (BigQuery AI.FORECAST nightly)	Re-forecasts the write-off on Approve; the nightly batch feeds Sense
Firestore	Stores practitioner feedback, live in the deployed service
Cloud Run, Jobs and Scheduler	Serves the app and API; runs Sense and Measure nightly; scales to zero when idle
MCP order tool	Places the customer's order when they accept the offer
Deterministic guardrails and estimator	Check every play against 8 rules and own every rupee; Gemini never sets a number

DEPLOY GitHub Actions runs verify, then deploys to Cloud Run on every merge to main.

ESTIMATED IMPLEMENTATION COST

Priced from measured usage; sold on rupees rescued.

PER 30 DAYS, MODELLED FROM MEASURED USAGE

₹486 / $5.86

Pre-switch (gemini-2.5-flash, 23 Sep). Not re-measured on gemini-3.5-flash.

Measured with no plays in the window: idle-plus-nightly cost.

Real Gemini tokens, 752 BigQuery jobs and Cloud Run time, priced at public list rates. Not a bill.

Method: modeled_from_measured_usage.

Forward-looking profiles ESTIMATES

Profile	Scale	Monthly
Micro	1 shop, ≤500 SKUs	₹200–500
Mid	300 SKUs, 10 nodes	₹2,500–5,000
Large	20,000 SKUs, 500 nodes	₹1–2 lakh

PLANNER PER PLAY · SEPARATE FIGURE, MODELLED

about ₹6.5 per play

gemini-3.5-flash, modelled at list prices, 14 Oct ($0.0777 at ₹83 per $)

BUYER AND PRICING

Buyer: the category or supply-chain head, who owns write-off.
Basis: a fee per node per month, or a percentage of rupees rescued vs the holdout.

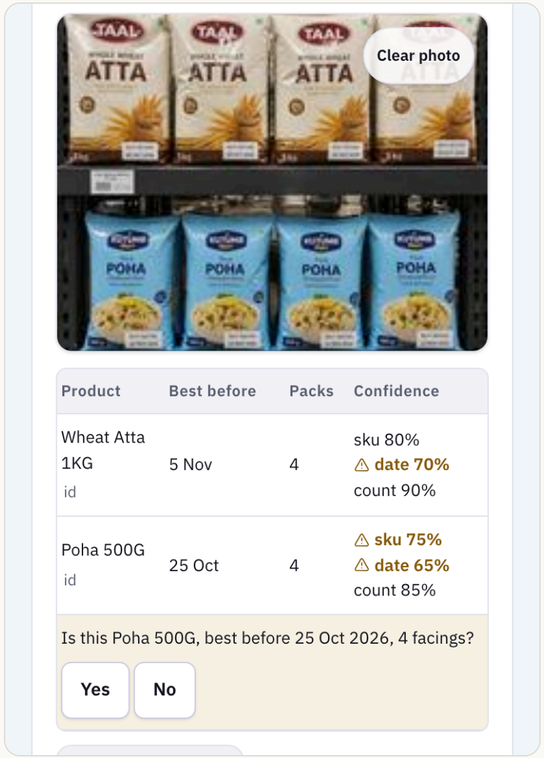
PROJECTED MARGIN LESS RUN COST ILLUSTRATION

≈ ₹21,710 – ₹156 to ₹313 per node

Per 28 days at the 5% prior: ₹347,356 margin over 16 nodes, less Mid-profile cost (₹2,500–5,000 a month, estimate). The ₹486 (pre-switch) had no plays in its window. Synthetic.

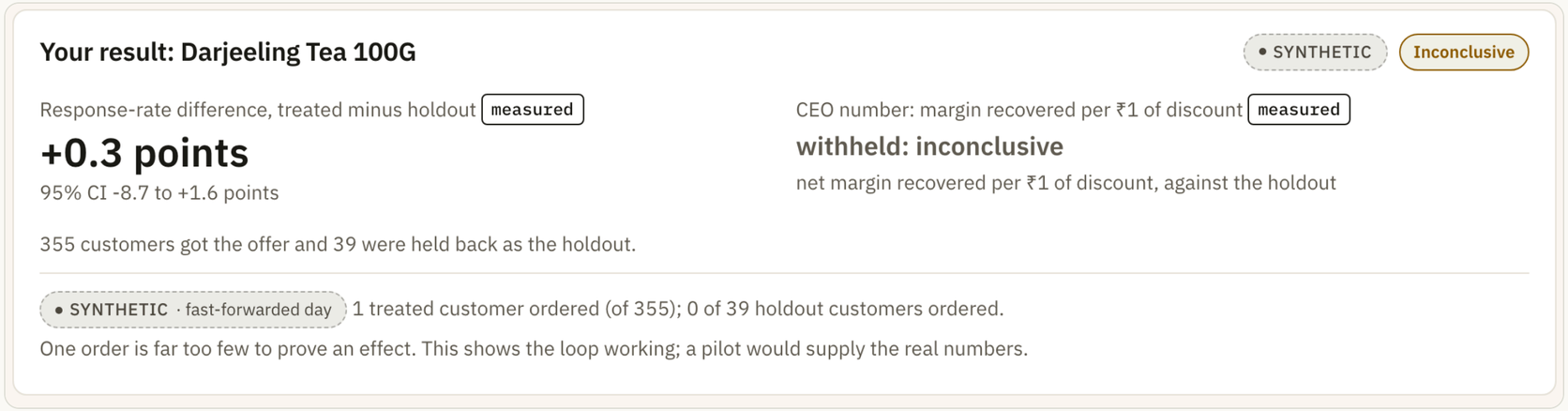
SNAPSHOTS OF THE PROTOTYPE

Doubtful reads go to a person; outcomes say SYNTHETIC.



Phone view

A read under 70% confidence is asked back before it reaches inventory.



Outcomes on a fast-forwarded day, synthetic orders: Tea

355 offered, 39 held out; 1 treated customer ordered, 0 held-out. Response rate +0.3 percentage points, 95% confidence interval –8.7 to +1.6. The range crosses zero, so the page says inconclusive and withholds the rupee figure.

PERFORMANCE REPORT

Measured and dated, on gemini-3.5-flash.

LIVE PLAN — 11 Oct

22/24

plans written by the model on the deployed app, 0 fallbacks

`plan_live_2026-10-12.json`

CUSTOMER CHAT — 14 Oct

3.66 s p50

p95 4.36 s, down from 8.44 s; Model Armor on

`customer_latency_2026-10-14_before_after`

APPROVE — 15 Oct

1.4–1.9 s

warm, ten samples, 13 to 15 Oct; 3.3 s before

`approve_timing_2026-10-15.json`

PHOTO INTAKE — 14 Oct

30/30

dates read correctly on synthetic photos, not real phone photos

`vision_synthetic.json`

ADVERSARIAL EVAL — 14 Oct

50 of 60 attack runs stopped

0 of 18 genuine requests refused. 10 of 60 runs (4 cases) got through; 3 cases fixed in code; V6 not fixed. Model Armor screens every chat message.

`eval/evaluation.md (adversarial)`

PLANNER EVALSETS — 14 Oct

39/46

first proposal accepted by the guardrails

`live_eval_2026-10-14/summary.json`

TESTS IN CI — every PR

1,700+

870 pytest + 833 Playwright (collected 16 Oct), in make verify before every deploy

`.github/workflows/verify.yml`

On the public backtest the forecaster does not beat a naive seasonal baseline (WAPE 1.12); Taal's value is the loop after the forecast (eval/external_backtest.md). Method and every source: eval/evaluation.md, deck/claims.md.

VALIDATION · PRACTITIONER RESPONSES, SEPTEMBER 2026

What seven practitioners told us.

3 of 7
write off stock weekly or daily

3 of 6
online sellers say the sell-by rule limits what they sell

0 of 7
measure a promotion against a control group

4 of 7
want a one-week pilot

"Blanket markdowns erode margin on products that may still sell normally."

Category / merchandising, quick-commerce

"Item-store recommendations with incremental-margin estimates."

Category / merchandising, quick-commerce

7 usable responses (10 collected, 3 excluded as flagged in config/feedback_exclusions.json). Anonymous by consent; contacts never published. Counts, not percentages.

USER GUIDE

Three people, three screens, one approved play.

Planner

/desk

1. Open the Play Desk and pick a gap.
2. Check the guardrails and counterfactuals.
3. Click Approve. The forecast moves.

Node manager

/phone

1. Photograph a pallet, or tap a sample.
2. Confirm the rows read below 0.7 confidence.
3. Approve the play from the gap card.

Customer

/chat

1. Open the chat. It starts with "Any offers today?".
2. Ask about a product. You see a stock band, not a count.
3. Send STOP to withdraw consent.

Badges. LIVE: a real call this visit. REPLAY: stored output replayed. SYNTHETIC: fabricated demo data. "Recorded from Gemini": a dated real model run.

IMPACT · WHAT IS MEASURED, WHAT IS PROJECTED

Projected margin hinges on one assumption.

TWO LOTS · PROJECTED

Tea (recorded Gemini run)

₹9,191 recovered

Vs doing nothing; projected, seeded tenant. ₹6,186 ahead of a blanket 20% markdown. 103 units, ₹35,020 at stake.

Chips (10 Oct, API snapshot)

-₹279

Projected. 368 units, ₹9,200 at stake. Net −₹8,468 vs −₹8,189.

RESPONSE PRIOR SENSITIVITY · PROJECTED

PRIOR	MARGIN	PER NODE / YEAR
2%	₹1.41 lakh	₹1.15 lakh
5% default	**₹3.47 lakh**	**₹2.83 lakh**
10%	₹6.19 lakh	₹5.04 lakh

Seeded tenant, 28-day run, 550 gaps. Per node / year = margin ÷ 16 nodes × 365/28.

MEASURED VS PROJECTED

Measured on real customers: none yet.

Synthetic holdout, Tea: 355 treated, 39 held back; 1 responder vs 0, inconclusive.

Power (SIMULATION): a 5 pp lift needs about 1,156 customers at a 90/10 split.

Projected at 5%: ₹347,356 margin, ₹706,183 waste avoided.

Practitioners: 7 usable responses; 3 of 7 write off stock weekly or daily.
Scale of the problem: sourced loss figure and arithmetic on slide 2 (as reported, all inventory loss, not a Taal result).

What it takes to run beyond the demo.

Production path

Today: services run as the owner-role deploy identity; CI uses a stored key. Secrets sit in a managed store. Fair-use limits cap plan, chat and photo calls per visitor and in total.

Next: least-privilege service accounts and workload identity.

Model IDs pinned with retirement dates: gemini-3.5-flash live, Flash-Lite fallback; the previous model retires 20 Oct 2026.

Scalability

Measured: Sense 3.8 s (local, 300 SKUs); chat p50 3.66 s, p95 4.36 s (14 Oct, gemini-3.5-flash, Model Armor on, local API); live plan p50 26.5 s, p95 31.8 s (22/24, 11 Oct); Approve 1.4 to 1.9 s warm (13 to 15 Oct; first call after idle takes longer).

Estimate: 20,000 SKUs, 500 nodes. ESTIMATE

Breaks first: max-instances 1 with re-plan state in memory, then the planner queue.

Feasibility

Buyer: category or supply-chain head. Fee per node or rupee share.

Pilot: 50 to 60 testers, one week, not yet run.

Planner: about ₹6.5 per play at list prices (gemini-3.5-flash, 14 Oct); offer copy generated once and stored.

Run cost: ₹486 a month, pre-switch (gemini-2.5-flash, 23 Sep), not a bill.

Estimates: ₹200 to 500 (Micro), ₹2,500 to 5,000 (Mid), ₹1 to 2 lakh (Large). ESTIMATE

CONNECTING TO A RETAILER The write-off owner (category or dark-store ops) pays. Stock with expiry and sales load from CSVs today; consented customers are a pilot's first step. Opted-in customers only, STOP in code, designed to India's DPDP consent model (not certified). Every play needs human approval.

ADDITIONAL DETAILS · FUTURE DEVELOPMENT

Next: a real pilot, real photos, more tenants.

1 A pre-registered pilot

One week, 50–60 consented testers, 40–50% holdout. Metric pre-registered in docs/pilot.md.

DESIGNED

2 Vision on real photos

Harness built. Real camera photos are the missing input; 30/30 (14 Oct) is on synthetic ones.

HARNESS BUILT

3 Multi-tenant BigQuery

Tenant id on every table; the BigQuery path is built, flag off until a second tenant.

BUILT, FLAG OFF

LEARNING CURVE · MEASURED, CHANGED, RE-MEASURED

MEASURED	BEFORE	WHAT WE CHANGED	AFTER
Chat p95, same 24 prompts	8.44 s	stock prefetch, Model Armor client reuse, minimal thinking	**4.36 s, Model Armor on (14 Oct)**
Approve, warm	3.3 s (12 Oct)	offer copy generated ahead of Approve	**1.4–1.9 s (13 to 15 Oct, ten samples)**
Live planner	no_play, 110.9 s (20 Sep, pre-switch)	gap lookup moved into code	**22 of 24 model-authored (11 Oct)**
Planner evalsets	43/46 match, 1/46 trajectory (old recordings)	re-recorded on gemini-3.5-flash	**46/46, 18/46: regression guard, not accuracy**
Adversarial eval	10 of 60 runs (4 cases) got through	code-only fixes	**3 cases fixed in code; V6 not fixed**

Rows are copied from docs/learning_curve.md, which also lists the changes that made a number worse. Pre-switch means gemini-2.5-flash.

LINKS

Taal: the agent that sells what the forecast says you'll throw away

DEMO VIDEO

drive.google.com/file/d/1hBKapntHMnc4FNybRqSEWods2MTrluBw

GITHUB REPOSITORY

https://github.com/Nandish3010/probable-fortnight

LIVE APP

https://taal-web-2obkp776ca-el.a.run.app

Transfer plus targeted nudge: ₹9,191 projected recovery vs doing nothing on one tea lot · seeded tenant

22 of 24 live plans on gemini-3.5-flash, p50 26.5 s (11 Oct)

Holdout so far: synthetic and inconclusive; a real pilot is next

Open the live app, click Run the 60-second beat, then Approve. Nothing you do there is saved.
Then, as Ravi, type: “Is the Darjeeling tea in stock?” · “Do you have Cola Zero?” · “STOP”

Team Taal Google Cloud AI Builder Cup 2026 · Retail & Commerce